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MILITARY SPECIFICATION

BREAD-SLICING MACHINES, ELECTRIC-OPERATED, MULTIPLE KNIFE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers electric-operated, multiple knife, bread-and slicing machines.

1.2 Classification. Bread-slicing machines shall be of the following types and styles as specified (see 6.2).

Type I - With gravity feed chute
Style 1 - With stand
Style 2 - Without stand

Type II - With mechanical single loaf feed (shipboard use)
Style 1 - With stand
Style 2 - Without stand

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7320

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SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 hp and Smaller)
QQ-N-288	- Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings
QQ-N-290	- Nickel Plating (Electrodeposited)
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting
MMM-A-260	- Adhesive, Water-Resistant, (For Sealing Waterproofed Paper)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-1055	- Barrier Material, Waterproofed Flexible
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-T-60	- Tape, Packaging, Waterproof
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Packaging/Paper (for Carton Sealing)

MILITARY

MIL-P-116	- Preservation, Methods Of
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment)

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement Of

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(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel, Plate, Sheet and Strip

D 3951 - Standard Practice for Commercial Packaging

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

BAKING INDUSTRY SANITATION STANDARDS COMMITTEE (BISSC)

Standard No. 13 - Bread and Roll Slicers and Wrappers

(Application for copies should be made to Baking Industry Sanitation Standards Committee, 521 Fifth Avenue, New York, NY 10017.)

UNDERWRITERS LABORATORIES, INC. (UL)

Standard No. 73 - Motor-Operated Appliance

(Application for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this document shall take precedence. Nothing in this document however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard product. The bread-slicing machine shall, as a minimum, be in accordance with the requirements of this document and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this document. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this document shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

* 3.2 Codes and standards. The bread-slicing machines shall conform to the applicable requirements of BISSC Standard No. 13, and UL Standard No. 73. Satisfactory evidence that these requirements have been met shall be submitted to the contracting officer or his authorized representative prior to the start of production (see 4.3).

3.3 Materials (see 6.3) and components. Materials and components shall be as specified herein. Materials and components not definitely specified shall be of the quality normally used by the manufacturer for his standard bread-slicing machine provided the completed items comply with all provisions of this document.

3.3.1 Stainless steel.

3.3.1.1 Stainless steel. Stainless steel shall conform to type 302 or 304 of ASTM A 167.

3.3.1.2 Nickel-copper alloy. Nickel-copper alloy shall conform to QQ-N-288.

3.4 Design and construction. The bread-slicing machines shall be electrically operated and be designed to have the knife frames rapidly reciprocated in opposite directions. Each loaf shall be completely sliced as it moves across the knives. A hold down unit shall be provided to hold the bread in its normal position during the slicing of the bread. Slices shall be cleanly cut and 7/16 inch in thickness. There shall be no marked change in the shape of the loaf after being sliced. The crumb tray shall be suitable for catching all crumbs that fall from the slicer. The crumb tray may be a pan or drawer. The machine shall be designed to transversely slice loaves, which are a minimum of 12 inches long and 4-5/8 inches square, at a rate of not less than twelve loaves per minute for type I and two loaves per minute minimum for type II when tested as specified in 4.4.2. The bread slicing machine shall consist of knives, knife frames (reciprocating powered type), loaf guides, crumb tray, back and front chute for type I, stand for style 1, motor, and control assembly. The type I bread-slicing machine shall have a chute equipped with loaf guides fixed at such an angle or angles that when the back chute is maintained fully loaded with loaves of bread,

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the loaves shall move by gravity to the knives, be sliced, and move by gravity to the bread stop at the front chute. The bread-slicing machine drives and mechanisms shall be enclosed to the maximum feasible to minimize potential for operator contact with moving parts. The type II machine shall have a single loaf receiving trough with hold down mechanism.

3.4.1 Knives. The slicing knives shall be heat-treated carbon steel having not less than 0.85 percent carbon with nickel or chromium plating except at the cutting edge, heat-treated stainless steel of cutlery grade having not less than 0.60 percent carbon and 14 percent chromium, or tempered and polished chrome-vanadium steel. The slicing knife cutting edge shall have a Rockwell hardness of C50 to C60 measured within 1/8 inch of the cutting edge when tested as specified in 4.4.3.

3.4.2 Knife frames. The knife frames shall be fabricated from aluminum, carbon steel, stainless steel, nickel-copper alloy, magnesium-base alloy, or malleable iron. The knife frames shall have the knives set to slice 7/16 inch thick slices. The frame shall be provided automatic means for independently adjusting the tension of each individual knife. The knife frame shall be constructed so as not to be deformed by loads arising from the tensioning of the knives.

3.4.3 Loaf guides, type I. Loaf guides shall consist of two full-length members, one on each side of the chute. The position of each member shall be readily and easily adjustable. The separation between the two members shall be adjustable from 6 inches to a maximum of 16 inches. Each member shall be provided with a mechanical clamping device or devices convenient for rapid manipulation by hand and securely locking the member in the desired position.

3.4.4 Crumb tray. The bread-slicing machine shall be provided with a crumb tray. The crumb tray shall be attached to the machine by mechanical means and shall be readily removable for cleaning.

3.4.5 Chutes, type I. The bread-slicing machine shall be provided with a back chute and a front chute. The back chute shall be at least 25-1/2 inches in length measured from the knives to the back edge of the back chute. The front chute shall be provided with a metal loaf stop and adjustable side guides. The front chute shall be at least 6-1/8 inches in length measured from the knives to the front edge of the front chute. Chutes shall be fabricated of stainless steel as specified in 3.3.1.1.

3.4.6 Stand (for style 1 only). The stand may be of a frame type or cabinet type construction. When frame type construction is used, the frame shall consist of four legs, top frame, and bottom frame. The top and bottom frames shall be welded or bolted to the four legs to form a rigid assembly. The legs shall extend from the floor to the bottom of the bread-slicing machine body. When a cabinet type construction is used, exterior sheets shall be welded or fastened to the frame to form a rigid assembly. The cabinet shall be provided with a door on one end. The door shall be provided with means to hold the door

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closed. Stands shall be provided with means for fastening the stand to the bread-slicing machine. The stands shall have holes for hold down bolts at the floor level.

3.4.7 Dimensions, type II. The dimensions of the type II machines without stands, but including all attachments necessary for operation, shall be 26 inches wide maximum, 29 inches deep maximum, and 28 inches high maximum.

3.5 Electric characteristics. Bread slicing machines shall be designed to operate on a nominal 120 volt, single phase 60 Hz power supply. Provisions for grounding shall be included in each slicer. Access to controls and connections shall be gained from the front of the slicing machine.

3.5.1 Motor. Unless otherwise specified (see 6.2), the motor for the bread-slicing machines shall be of the totally inclosed type. The motor shall have the necessary characteristics to start and operate the machine under full-load conditions without exceeding the temperature for which the motor was designed. The motor shall be equipped with external overload protection. Manual reset overload protection will be accepted if this is a common practice of the manufacturer. The motor shall conform to CC-M-1807.

3.5.2 Switch motor. The bread-slicing machine shall have a readily accessible double-pole, single-throw switch provided for starting and stopping the motor. The switch shall be recessed into the body structure of the bread-slicing machine or may be shielded to prevent inadvertent activation.

3.5.3 Cord and attachment plug. All bread-slicing machines shall be furnished with a flexible, oil-resistant, rubber covered, type SO or SJO cord with a 3-prong grounding type plug. The cord shall be 3-wire grounding type and shall extend not less than 6 feet beyond the machine.

3.6 Operation. The bread-slicing machine shall operate properly and shall not bind, adjustable components shall be adjustable over the entire range without binding, and starting and stopping devices shall operate properly when tested as specified in 4.4.1. After operation there shall be no loose, broken, or missing parts and no wear of moving components.

3.7 Electromagnetic compatibility. When specified (see 6.2), the units shall be designed and equipped for electromagnetic compatibility in accordance with class V, class C3, part 10, group I of MIL-STD-461 (see 4.4.4).

3.8 Fungus and moisture resistance of electrical components. When specified (see 6.2), bread-slicing machines shall be treated to resist fungus and moisture as follows:

a. Circuit elements which have a temperature rise of not more than 75°F when operated at full load, shall be coated with fungus-resistant varnish conforming to MIL-V-173. Circuit elements include, but are not limited to, cable, wire, terminal blocks, capacitors, and coils.

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b. Electrical components not requiring fungus moisture resistance such as switches, fuses, and contacts, shall not be treated. Other materials and components which are inherently fungus-resistant or are protected by hermetic sealing need not be treated.

3.9 Lubrication. Bearings used for rotating members, except permanent lubricated bearings, shall have oil cup, grease cup, or pressure injection type grease fittings, except the ball or roller type bearings may be grease packed and sealed, and sleeve bearings may be lubricated in connection with an oil-filled housing of which they are a part. All other bearings, except permanent bearings, used for guiding or supporting the frames, shall be accessible for lubrication and shall be so arranged that no oil or grease can come in contact with the bread.

3.9.1 Tools. If lubrication fittings of the grease-gun injection type are used, a hand-operated grease-gun shall be supplied with the machine. Each bread-slicing machine shall be provided with a knife removing tool.

3.10 Navy shipboard installation (type II). The type II bread-slicing machine shall conform to the additional requirements in 3.10.1 and 3.10.2.

3.10.1 Mounting legs. The bottoms of the legs of the bread-slicing machine frame shall be provided with holes for mounting. Four holes shall be provided each threaded to receive a 3/8 - 16 minimum hold down bolt.

3.10.2 Inclined operation. Slicing machines shall operate satisfactorily when inclined at an angle of 15 degrees each side of the vertical in each of the two vertical planes at right angles to each other, with no spillage of product when tested as specified in 4.4.2.

3.11 Finish. Exterior surfaces and parts not specified shall be neatly finished with a durable enamel in accordance with standard commercial practice or may be nickel-copper alloy, stainless steel, or an aluminum-base alloy having an electrolytically oxidized finish. The interior of the knifeguard surfaces shall be enameled. Exterior surfaces or parts of nickel-copper alloy and stainless steel shall be smooth and polished. No cadmium plating shall be used.

3.11.1 Chromium plating. When chromium plating is used, the plating shall be copper, nickel, and chromium deposited in successive layers of such sequence that, after buffing and polishing operations, the outermost layer is chromium not less than 0.00001 inch thick, the layer immediately below the chromium is nickel not less than 0.00060 inch thick and the total thickness of all layers is not less than 0.00150 inch.

3.11.2 Nickel plating. Nickel plating shall conform to the requirements of class 1, type I (DS) of QQ-N-290.

3.12 Data-name plates. The slicing machines shall be furnished with a data-name plate in accordance with the applicable requirements of MIL-STD-130 except

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the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required and (d) Optional marking information shall not apply. The data plates shall be made of minimum 20-gauge corrosion-resisting metal and attached to the slicing machines by rivets, screws, or welding in such a manner as to meet the applicable National Sanitation Foundation sanitary requirements for this equipment. The plate shall contain the following information which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number and Serial Number
Government Approved Manual Number (see 6.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the slicing machine.

3.13 Workmanship. All components and assemblies of the bread-slicing machine shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts, and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and sharp corners, except when corners are required.

3.13.1 Welding or soldering. The surfaces of parts to be welded or soldered, as applicable, shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts, or incompleted fusion. Soldering shall be complete, clean, adherent, and without pinholes. All scale and flux deposits (when flux is used) shall be removed from the finished weld area.

3.13.2 Fastening devices. Threaded fastener and rivet holes shall be free of burrs. Threaded fasteners shall not be broken, cracked, or stripped. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible or the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such

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inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine validity of the certification.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.1.1 Component testing. The slicing knives shall be tested as specified in 4.4.3 to determine compliance with 3.4.1. The sample unit shall be one knife. The inspection lot shall be all the knives submitted for inspection at one time. The inspection level shall be S-2. Any test failure shall be cause for rejection of the lot.

4.2.2 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of bread-slicing machines of one type and style offered for acceptance at one time. The sample unit shall be one bread-slicing machine. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Standard product	Item not in accordance with manufacturer's standard product except for any added features needed to comply with document requirements	X	
Finish	Type of finish not as specified or blistered, peeled, chipped, or area of no film or thin film	X	
Design	Characteristic not in accordance with the specified requirements	X	
Construction and workmanship (general)	Any component missing or not as specified	X	
	Any visible component fractured, split, or malformed	X	
	Adjustable assembly that cannot be adjusted	X	
	Adjustable assembly that cannot be adjusted to perform the function intended		X
	Casting miscast, cracked, porous, or contains blowholes	X	
	Assembly or component improperly located (for example, gasket improperly fitted to provide seal)	X	
	Functioning components that require abnormal force to operate	X	
	Crack or crevice not filled and sealed to prevent entrance of dust, liquid, and dirt	X	
	Floor stand not provided with holes for holddown bolts	X	
	Crumb tray not attached to the machine by mechanical means	X	
	Crumb tray not readily removable for cleaning	X	
	Components not readily accessible for servicing where required		X
	Any sharp edge, corner, sliver, burr, rough die and grind marks, dent, or crack	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Threaded fasteners	Threaded fasteners missing, broken, stripped, fractured, or loose	X	
Welding and brazing (where required)	Not type specified, missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or flux deposits not removed		X
Tools	Missing		X
Data-name plate	Omitted or not as specified		X
	Information incomplete or not legible		X
	Not located so as to be readily visible to operator		X

4.2.3 End item dimensional examination. The end item shall be examined for compliance with the dimensions specified in 3.4.3, 3.4.5 and 3.4.7. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of bread-slicing machines of one type and style offered for acceptance at one time. The sample unit shall be one complete unit. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.2.4 End item testing.

4.2.4.1 Operational testing. Every end item shall be tested as specified in 4.4.1. Any end item exhibiting one or more test failures shall be rejected.

4.2.4.2 Representative item testing. One representative end item from the lot shall be tested as specified in 4.4.2 and, when applicable, 4.4.4. Any test failure shall be cause for rejection of the lot.

4.2.5 Packaging examination. An examination shall be made to determine that the preservation, packing, and marking comply with section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Markings (exterior and interior container)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Materials	Any component missing, damaged, or otherwise defective.
Workmanship	Items not disassembled, when applicable. Contents of interior packages not immobilized as specified. Bulged or distorted container.

4.3 Code and standard compliance. Certifications, certified test reports or listing marks for standards submitted in accordance with 3.2 shall be examined and validated as proof of compliance. Prior to Government approval of first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the bread slicing machines he proposes to furnish under this document meet the applicable requirements of BISSC, and UL as follows:

4.3.1 BISSC. Acceptable evidence of meeting requirements of BISSC Standard No. 13 shall be:

a. A photostatic copy of the current BISSC certificate of authorization for bread-slicing machine, or

b. A certified test report, acceptable to the contracting officer with the advice of the Army Surgeon General from an independent testing laboratory, indicating that the bread slicing machine has been tested and conforms to the specified BISSC Standard.

4.3.2 UL certification. Acceptable evidence of meeting the applicable requirements of UL Standard No. 73 shall be a UL listing mark or a certified test report from a recognized independent testing laboratory, acceptable to the Government indicating that the bread-slicing machine has been tested and conforms to the applicable requirements of UL Standard No. 73.

4.4 Methods of inspection.

4.4.1 Operational test. Connect the bread-slicing machine to an appropriate electrical power source and operate the unit for a minimum of 2 hours to determine compliance with the requirements specified in 3.6. While operating, start and stop the machine at least five times, observe bearings for overheating, observe starting and stopping devices for proper operation, observe moving components for binding or improper operation, and also move adjustable

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components over the range of adjustment and observe for binding or improper operation. When power is turned "OFF" check for conformance with 3.6. Any nonconformance with the requirements specified shall constitute failure of this test.

4.4.2 Functional test. After test in 4.4.1 is completed, the unit shall be tested with a load for 5 minutes to determine that the machine slices bread at the rate specified in 3.4, and 3.10.3 when applicable. The shape of the loaf shall be examined before and after slicing and the slice thickness and surfaces shall be examined to determine compliance with the requirements specified in 3.4, and 3.10.2 when applicable. Any nonconformance with the requirements specified shall constitute failure of this test.

4.4.3 Knife hardness test. The hardness of the slicing knives shall be determined within 1/8 inch of the cutting edge in accordance with ASTM E 18. Three hardness determinations shall be made on each sample unit and the average of the three determinations shall be recorded. Any sample unit with an average hardness not falling within the range specified in 3.4.1 shall be considered a test failure.

4.4.4 Electromagnetic compatibility tests. When electromagnetic compatibility is required, the test unit shall be tested by the contractor in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the contractor or an independent testing agency. The contractor shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the Electromagnetic Compatibility Test Report and the requirements meet MIL-STD-461. Any nonconformance with the requirements specified shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2). The extent of disassembly shall be determined by the manufacturers. Parts removed shall be adequately identified or tagged. Disassembly shall not be such that special skills or special tools are required to place that equipment in operation.

5.1.1 Level A.

5.1.1.1 Disassembly. Projecting detachable components shall be detached from each bread-slicing machine in order to reduce cube and prevent damage while in transit.

5.1.1.2 Cleaning. All exposed, uncoated, ferrous metal surfaces of each machine, including the slicing knives, detached parts, and tools, shall be cleaned by process C-1 of MIL-P-116 and thoroughly dried.

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5.1.1.3 Application of preservative. Surfaces, cleaned and dried as specified, shall be coated with type P-14 preservative of MIL-P-116.

5.1.1.4 Protection of preservative. Surfaces, from which preservative might be removed by contact with packaging materials, shall be covered, wrapped, or bagged with greaseproof barrier material conforming to type I and II, grade A, class 1 or 2 of MIL-B-121. Barrier material shall be secured with twine made of jute, sisal, manila, or cotton, or with tape conforming to type III, grade A of PPP-T-45, PPP-T-60 or PPP-T-76, or heat sealed when heat-sealable barrier material is used.

5.1.1.5 Belts and pulleys. Uncoated ferrous metal surfaces of pulleys shall be coated with primer conforming to TT-P-664 and the primer dried. Belts shall be replaced and tension relieved.

5.1.1.6 Unit packing. Components of the machine that are free to move shall be secured in such a manner as to prevent movement or vibration while in transit. Detached parts and tools shall be unit packed in a snug-fitting fiberboard box or boxes conforming to style optional class weather-resistant of PPP-B-636. Contents of each box shall be cushioned and immobilized, as necessary, with fiberboard inserts or other suitable means in such a manner as to prevent damage while in transit. Each box shall be securely closed in accordance with method IV as specified in the appendix of PPP-B-636.

5.1.2 Commercial. Bread-slicing machines shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Each bread-slicing machine, preserved as specified in 5.1, shall be packed in a nailed-wood or cleated-plywood shipping container conforming to class 2, style 2 or 4, grade A or B, type 3 load of PPP-B-621, or overseas type, style A or J, grade A or B, type 3 load of PPP-B-601, respectively. The machine shall be secured by carriage bolts passing through the skids or through load bearing member attached to the inside surface of the bottom of the container, or by lumber blocking and bracing. Blocking and bracing shall not depend on end-grain nailing alone, but supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing. The machine shall be covered with a full-height shroud made of waterproof barrier material conforming to class E-2, H-5, or M-1 of PPP-B-1055. All seams of the shroud shall be sealed with 3/4 inch minimum width continuous seam of water-resistant adhesive conforming to MMM-A-260. The fiberboard box containing detached components and tools shall be secured in an unused portion of the shipping container. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container document.

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5.2.2 Level B packing. Each bread-slicing machine, preserved as specified in 5.1, shall be packed in a snug-fitting shipping container conforming to domestic type, style A or J, grade B, type 3 load of PPP-B-601 or class 1, style 2 or 4, grade B, type 3 load of PPP-B-621. The machine shall be secured, blocked and braced as specified in 5.2.1. The fiberboard box containing detached components and tools shall be secured in an unused portion of the shipping container. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container document.

5.2.3 Commercial packing. Bread-slicing machines, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The type I bread-slicing machine is intended for use in fixed baking installations. The type II bread-slicing machine is intended for use by the Navy for shipboard installation.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type, and style of bread-slicing machine required (see 1.2).
c. When motor is not to be totally inclosed type (see 3.5.1).
d. When electromagnetic compatibility is required (see 3.7).
e. When fungus and moisture treatment of electrical components is required (see 3.8).
f. Government approved manual number to be included on data-name plate (see 3.12).
g. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

6.4 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

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Navy - YD
Air Force - 99

Review activities:

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Navy - MC
Air Force - 84
DLA - GS

User activity:

Navy - SA

Preparing activity:

Army - GL

Project No. 7320-0761

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-43866C	BREAD-SLICING MACHINES, ELECTRIC-OPERATED, MULTIPLE KNIFE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP-1, Feb 86

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